

Rafael F Barduzzi

PhD candidate

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Taxonomy and Evolution of Plants lab
Department of Botany
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EDUCATION

- 2024-present **Ph.D. in Comparative Biology**, Universidade de São Paulo
Title: Unravelling the tempo and mode of independent radiations in the megadiverse genus *Mimosa* L. (Fabaceae, Mimoseae)
Dissertation advisor: Dr. Leonardo M. Borges
Research supervisor (Abroad): Dr. Daniel S. Moen
- 2018-2023 **B.Sc. in Biology**, Universidade Federal de São Carlos
Undergraduate research: Pollen disparity in the ‘Piptadenia group’ (Leguminosae): differences between lineages and ecological groups
Advisor: Dr. Leonardo M. Borges

APPOINTMENTS

- 2025 **Visiting Scholar**, Mississippi State University
- 2025 **Research Scholar**, University of California Riverside
- 2024-present **Graduate Researcher**, Universidade de São Paulo
- 2021-2023 **Undergraduate Researcher**, Universidade Federal de São Carlos

SCHOLARSHIPS

- 2025 São Paulo Research Foundation (FAPESP) Research Internship Abroad Scholarship
- 2024-present FAPESP Doctoral degree (Direct) Scholarship
- 2021-2023 FAPESP Undergraduate Research Scholarship

PUBLICATIONS

Barros-Souza, Y., Maianne, M., **Barduzzi, R. F.**, & Borges, L. M. (2026). Beyond the Highlands: Climate Drives Evolutionary Connections Between Ancient Neotropical Mountains and

Lowland Biomes. *Global Ecology and Biogeography*, 35(2), e70214. <https://doi.org/10.1111/geb.70214>

Barduzzi, R. F., Liau-Kang, S., Duarte, A. F. T., Santos, F. D. A. R. D., & Borges, L. M. (2026). Morphological innovation and lineage-specific history drive disparification in the aggregated pollen of mimosoid plants. *Annals of Botany*, 137(1), 159-170 <https://doi.org/10.1093/aob/mcaf213>

Barduzzi, R. F., Barros-Souza, Y., Liau-Kang, S., & Borges, L. M. (2024). Pollen evolution in the Stryphnodendron clade (Leguminosae, Mimoseae). *Brazilian Journal of Botany*, 48(1), 15. <https://doi.org/10.1007/s40415-024-01059-y> (Part of special issue, *Advances in Legume Systematics* 15)

Liau-Kang, S., **Fernandes Barduzzi, R.**, Lozano Calderón, G., Marega Junior, E., Simon, M. F., Ribeiro dos Santos, F. D. A., & Borges, L. M. (2024). More of almost the same: novel pollen descriptions uncover subtle morphological variation in *Mimosa* L. (Leguminosae, Mimoseae). *Brazilian Journal of Botany*, 47(4), 1071-1090. <https://doi.org/10.1007/s40415-024-01033-8> (Part of special issue, *Advances in Legume Systematics* 15)

MANUSCRIPTS SUBMITTED

Duarte, A. F. T., **Barduzzi, R. F.**, Ribeiro, P. G., Queiroz, L. P., & Santos, F. A. R. *In Press*. Morphological characterization of the pollen grains of species from the new genus *Marlimorimia* and the new circumscription of *Pityrocarpa*.

Barduzzi, R. F., Santos, M., & Borges, L. M. *Under review*. Tree thinking and alternative knowledge among undergraduate Biological Sciences students.

PRESENTATIONS

Talks

- 2026 **Barduzzi, R. F.**; Moen, D. S.; Barros-Souza, Y.; Borges, L. M. A radiation space for *Mimosa*: Evolutionary radiations underlie diversity of one of Brazil's most species-rich plant genera. 2nd Brazilian Congress of Evolutionary Biology, Belo Horizonte (MG), Brazil
- 2026 **Barduzzi, R. F.** Toward a radiation space of the sensitive plants. V Symposium on Comparative Biology, Ribeirão Preto (SP), Brazil
- 2025 **Barduzzi, R. F.** Evolutionary radiations in the genus *Mimosa* L. (Leguminosae). UCR Center for Conservation Biology's 7th annual EcoSlam, Riverside (CA), USA

- 2024 **Barduzzi, R. F.** How to identify evolutionary radiations. 11th Meeting on Comparative Biology, Ribeirão Preto (SP), Brazil
- 2023 **Barduzzi, R. F.;** Liao-Kang, S.; Santos, F. A. R. and Borges, L. M. Pollen disparity in the *Mimosa* and *Stryphnodendron* clades (Caesalpinioideae, Mimosoid clade): comparing lineages and ecological groups. 8th International Legume Conference, Pirenópolis (GO), Brazil
- 2022 **Barduzzi, R. F.** and Borges, L. M. Pollen disparity in *Piptadenia* group (Leguminosae), 2022, São Carlos. 28th Universidade Federal de São Carlos Congress on Undergrad Research, São Carlos (SP), Brazil

Posters

- 2025 **Barduzzi, R. F.;** Moen, D. S.; Barros-Souza, Y.; Maianne, M.; Borges, L. The role of leaf morphology, elevation, soil and climate in *Mimosa* L. (Leguminosae) diversification. Botany 2025, Palm Springs (CA), USA
- 2024 **Barduzzi, R. F.;** Barros-Souza, Y.; Maianne, M.; Borges, L. M. Radiations of the sensitive plants: exploring the influence of abiotic niche in diversification of the mega-genus *Mimosa* L. (Leguminosae). 1st Brazilian Congress of Evolutionary Biology, Curitiba (PR), Brazil
- 2023 **Barduzzi, R. F.;** Liao-Kang, S.; Barros-Souza, Y.; Borges, L. M. Evolution of pollen morphology in the *Stryphnodendron* clade (Fabaceae, Caesalpinioideae). 73th Brazilian Congress of Botany, Belém (PA), Brazil
- 2022 **Barduzzi, R. F.;** Liao-Kang, S. and Borges, L. M. Pollen disparity in the *Piptadenia* group (Leguminosae): differences between lineages and ecological groups. 2022. 72th Brazilian Congress of Botany, Brazil

TEACHING EXPERIENCE

Academia

- 2026 Pollen grains: Understanding their morphology and importance for taxonomy — 2nd edition, **Short course** at the XVIII Universidade Federal de São Carlos Biology Week (*Semana Acadêmica da Biologia*), Lead organizer
- 2025 Phytogeographic Analyses in R, **Short course** at the IV Rio–São Paulo Botany Meeting (*Jornada Rio–São Paulo de Botânica*), co-organizer.

- 2024 Pollen grains: Understanding their morphology and importance for taxonomy — 1st edition, **Workshop** at the XVI Universidade Federal de São Carlos Biology Week (*Semana Acadêmica da Biologia*), Lead organizer
- 2023 Behavioral Biology, **Undergraduate course**, Universidade Federal de São Carlos, Teaching Assistant

Student supervision

- 2026-present Diogo Mantovanelli Rabetti (Universidade Federal de São Carlos – **Undergraduate Research**). Co-supervision with Leonardo M. Borges
- 2025-2026 Felipe Ribeiro Alves Selotto (Universidade Federal de São Carlos - **Bachelor's Thesis**). Co-supervision with Leonardo M. Borges

PROFESSIONAL AFFILIATIONS

Society for the Study of Evolution (2024–)
Brazilian Society of Evolutionary Biology (2024–)

ACADEMIC SERVICE

Student Representative, Undergraduate Program Council in Biological Sciences (2019-2023). Represented undergraduate students in curricular and academic matters and participated in program council meetings and discussions.

LANGUAGES

Portuguese (native); English (fluent)

FIELDWORK EXPERIENCE

Central Brazil (2023); Southwest Brazil (2024, 2026). Collected and prepared botanical specimens for herbarium deposition and collected data on plant–pollinator interactions.